Huawei Access Node (BAS-ACC-05) QoS Congestion Event Resolution SOP (ALARM_6215)

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6215

Description

This Standard Operating Procedure addresses the failure domain of QoS congestion events, manifesting as microbursts, interface tail drops, and QoS queue starvation on the Huawei Access Node, BAS-ACC-05, located at the Basavanagudi POP. The underlying cause of this failure domain is attributed to the interaction between the node's Quality of Service (QoS) mechanisms and the network's traffic patterns. Specifically, the Huawei Access Node utilizes the Multi-Protocol Label Switching (MPLS) Label Edge Router (LER) function to provide differentiated services and traffic prioritization, which, when not properly configured or managed, can lead to congestion events. Furthermore, the node's hardware, including the forwarding engine and buffer resources, can contribute to tail drop events when the interface congestion exceeds the buffer capacity. To address this failure domain, it is essential to analyze the QoS configuration, verify the MPLS LER function, and inspect the hardware resources to identify the root cause of the congestion event.

Pre-requisite Checks

- prerequisite: Verify redundant link status and confirm that the BAS-ACC-05 node is not experiencing link failures or flapping.

- prerequisite: Check the environmental conditions at the Basavanagudi POP, including temperature, humidity, and power supply levels, to ensure that they are within the recommended operating ranges.

- prerequisite: Gather historical traffic data and QoS configuration information from the BAS-ACC-05 node to aid in the analysis of the congestion event.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies or profiles	High
MPLS LER function misconfiguration or malfunction	Medium
Hardware resource limitations (buffer size, forwarding engine capacity)	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy-profile, Expected Output: Policy Profile Name: P1, Class Name: C1, Action: set-traffic-policing cir 1000000, Description: Verifies the QoS policy profiles and class configurations to identify any potential misconfigurations or anomalies.

Command: display mpls l2circuit, Expected Output: MPLS L2 Circuit Information: L2 Circuit ID: 1, L2 Circuit Type: L2VPN, State: UP,
```

```
Description: Inspects the MPLS L2 circuit information to verify the
L2VPN configuration and ensure it is in the UP state.

Command: display interface, Expected Output: Interface Information:
Interface Name: Eth-Trunk1, Interface Status: UP, MTU: 1500,
Description: Analyzes the interface information to verify the link
status, MTU, and bandwidth allocation.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy profiles or class configurations, Command: system-view qos policy-profile P1 class C1 traffic-policing cir 1000000, Description: Reconfigures or updates the QoS policy profiles or class configurations to address any misconfigurations or anomalies identified during the diagnostic process.

Step 2: Step: Verify and adjust MPLS LER function, Command: system-view mpls l2circuit L2Circuit-1 modify circuit-type L2VPN modify state UP, Description: Reconfigures or updates the MPLS LER function to ensure it is correctly configured and functioning as expected.

Step 3: Step: Verify and adjust hardware resources, Command: system-view interface Eth-Trunk1 modify mtu 1600 modify bandwidth 1000, Description: Reconfigures or updates the hardware resources, including the MTU and bandwidth allocation, to address any limitations or deficiencies identified during the diagnostic process.

Post-Check Verification

```
Command: display qos policy-profile, Expected Output: Policy Profile
Name: P1, Class Name: C1, Action: set-traffic-policing cir 1000000,
Description: Verifies that the QoS policy profiles and class
configurations have been successfully updated and are functioning as
expected.

Command: display mpls l2circuit, Expected Output: MPLS L2 Circuit
Information: L2 Circuit ID: 1, L2 Circuit Type: L2VPN, State: UP,
Description: Verifies that the MPLS L2 circuit information has been
successfully updated and is functioning as expected.

Command: display interface, Expected Output: Interface Information:
Interface Name: Eth-Trunk1, Interface Status: UP, MTU: 1600,
Description: Verifies that the interface information has been
successfully updated and is functioning as expected.
```

Roll-back Procedures

- Step: Restore original QoS policy profiles or class configurations, Command: system-view undo qos policy-profile P1 undo class C1 undo traffic-policing cir 1000000, Description: Restores the original QoS policy profiles or class configurations to their previous state in the event of a failed fix or rollback.

- Step: Restore original MPLS LER function, Command: system-view undo mpls l2circuit L2Circuit-1 undo modify circuit-type L2VPN undo modify state UP, Description: Restores the original MPLS LER function to its previous state in the event of a failed fix or rollback.

- Step: Restore original hardware resources, Command: system-view undo interface Eth-Trunk1 undo modify mtu 1600 undo modify bandwidth 1000, Description: Restores the original hardware resources, including the MTU and bandwidth allocation, to their previous state in the event of a failed fix or rollback.